

Lucas Picard

Double Major: [Mechanical & Robotics Engineering](#) | [Mechatronics](#) · [CAD](#) · [Embedded Systems](#) · [Prototyping](#)
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SUMMARY

I am a mechanical engineering student at WPI (sophomore standing, 3.75 GPA) pursuing a double major in Mechanical and Robotics Engineering, captivated by the challenge of building things others think are impossible. I design and bring real hardware to life — embedded electronics, robotic systems, and CAD assemblies — and write the software and automation that drives them. Right now I am automating digital engineering processes as an engineering intern at a contract electronics manufacturer, and as a dual US/German citizen, bilingual in English and German, I am driven to push the boundaries of what is possible in the STEM field.

EXPERIENCE

Microboard Processing — *Engineering Intern, Document & Controls* June 2026 – Present
Seymour, CT — contract electronics manufacturer (ITAR-regulated).

- Automate digital engineering and documentation processes alongside an industrial engineer, streamlining manufacturing workflows.
- Built standardized SMT/PCB assembly process-flow documentation that sharpened consistency and clarity across the production line.
- Apply disciplined data-handling and documentation practices within a regulated, compliance-sensitive environment.

Picard Studio — *Studio Assistant & Operations* Feb 2020 – Present
Southbury, CT — art education business (online courses & membership).

- Built and maintained KPI dashboards tracking MRR, ARPU, retention, churn, and acquisition to drive business decisions.
- Automated monthly metrics reporting through the Kajabi API and GitHub Actions, piping data straight into Google Sheets.
- Authored a 459-record membership cancellation analysis (13 charts, 15-page report) that surfaced renewal-window and engagement-based retention signals.
- Delivered freelance CAD work — designing, prototyping, and invoicing a wall-mount equipment cradle.

Aquila's Nest Vineyards — *Service Staff (Seasonal)* Jul – Nov 2024

New Morning Market — *Grocery Clerk* Aug – Nov 2023

EDUCATION

Worcester Polytechnic Institute — *B.S. Mechanical Engineering & Robotics Engineering (Double Major)*
Expected 2029

Worcester, MA — Sophomore standing; entered with 21 AP transfer credits (Calculus I–II, Physics C Mechanics & E&M, Computer Science A). GPA: 3.75 / 4.00. Dean's List.

Completed coursework: Statics, Dynamics, Thermodynamics, Manufacturing Science & CNC Machining, Introduction to Robotics, Intro to ECE, Calculus III–IV, Differential Equations, Linear Algebra.

Upcoming: Unified Robotics I–IV, Fluid Mechanics, Heat Transfer, Stress Analysis, Control Engineering, Materials Science, Software Engineering.

Activities: Combat Robotics, Club Badminton, Intramural Soccer.

Pomperaug High School — *Southbury, CT* 2021 – 2025
GPA: 4.06 / 4.5 scale. Honor Roll, all years. German Seal of Biliteracy.

ENGINEERING PROJECTS

SCARA Robot Arm — *Personal mechatronics build* 2026 – Present

- Repurposing salvaged Anet A8 3D-printer components (NEMA 17 steppers, frame hardware) into a 4-axis SCARA robot arm.
- Integrated AS5600 magnetic encoders for closed-loop joint feedback, with custom-designed, 3D-printed arm segments that house dedicated encoder mounts.
- Engineered the drivetrain around a Tr8x2 lead screw and closed-loop GT2 belts; assembled the J1 base and am now driving it through motion testing.

ESP32-P4 AI Camera Pen — *Embedded hardware / PCB design* 2025 – Present

- Designing a 12 mm-diameter handwriting-capture pen on a dual-PCB stack with an OV5640 camera and 0.95" AMOLED display.
- Captures handwriting, processes it through cloud AI over WiFi, and brings the results to life on-device — KiCad schematic complete, PCBs laid out (~65%).

CT Used-Car Scraper — *Python automation tool* 2026

- Built a Python + Playwright web scraper that aggregates used-car listings across multiple Connecticut dealerships, filtering by make, model, price, and mileage to surface the best candidates.

FRC Robotics — *Team Captain & Lead Driver · 1000+ hours* 2021 – 2025

- Led CAD, mechanical build, and electrical integration across subteams, designing robot subsystems in Onshape / SolidWorks and troubleshooting rapidly in the pit.
- As primary driver, earned the team's first New England District Championship qualification in 10 years and its first CT State Championship (off-season).

TECHNICAL SKILLS

CAD & Design: SolidWorks, Onshape, GD&T, design for manufacturability

Electronics: KiCad PCB design, soldering, embedded systems (ESP32 / Arduino), closed-loop control, sensors & encoders

Prototyping: 3D printing, manual milling, laser cutting, CAM

Software & Data: Python, Playwright, REST API automation, GitHub Actions, Excel / Google Sheets KPI analysis

Other: Technical documentation, ITAR-aware data handling

AWARDS · LANGUAGES · INTERESTS

Awards: Dean's List (WPI, Fall 2025); Honor Roll (Pomperaug, all years).

Languages: English (native), German (Seal of Biliteracy) — dual US/German citizen.

Interests: Acrylic painting, drawing, music performance, robotics.